KUO, CHEN CHIH 郭晨志

M.S. in Chemical Engineering • Licensed environmental specialist

Applying for **ISEP EHS Engineer**

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Xinzhuang, New Taipei · Open to relocation / shifts
Other Military-service exempt · Car licence · Graduating Aug 2027

Profile

NTUST Chemical Engineering M.S. candidate holding three environmental specialist licences (**wastewater, waste, indoor air**), OSH exams scheduled (Level B Nov 2026, Level A Mar 2027); hands-on with **hazardous materials** (lithium metal, organic-solvent electrolytes, faulty Li–$SOCl_2$ cells), lab safety and **thermal hazard analysis (TGA / DSC)**, plus **dry-room** clean-area work. Seeing synchrotron equipment run for 30+ years on preventive maintenance convinced me that systems are what keep people safe.

Experience

2025 – Present — **Sustainable Electrochemical Energy Development (SEED) Center, NTUST** • MOE Featured Areas Research Center

Lab Equipment, Asset Maintenance & Safety Officer

- **Procurement (>NTD 2M):** sourced, accepted and logged safety-critical equipment such as glovebox O_2 sensors, plus annual service contracts and consumables.
- **Lab safety:** monitor glovebox O_2 / moisture levels and handle sensor faults; run daily gas-cylinder checks and lab safety walk-throughs and brief newcomers on safety; led the on-site response when a broken bottle injured a student — sent one person with the injured student, one to alert the lab lead and asked for the SDS to be checked, then absorbed the spill, cleared the glass and ventilated the area; also first on scene for a glovebox catalyst-canister leak. Afterwards I audited the lab's protective equipment, found expired respirator cartridges and Hexafluorine, and bought replacements.
- **Hazardous materials:** handle lithium metal and assemble coin cells in a glovebox, prepare ether / carbonate organic-solvent electrolytes, and dismantle and handle faulty Li–$SOCl_2$ cells.
- **TGA testing service:** Ran TGA measurements and data interpretation for graduate students from 6 labs across Chemical, Mechanical and Materials Engineering at NTUST — over 600 samples in total, including my own lab's.
- **Rules:** wrote the TGA's PPE and ventilation rules and the TGA / DSC booking and usage rules that every user, including other labs, must follow — e.g. no DSC heating of unknowns without prior TGA decomposition data; volatile solvents only in pressure-rated pans.
- **Instrument maintenance:** Responsible for maintenance and calibration of the key thermal analysis instruments, TGA Q500 and DSC 2500.

Summer 2022 — **New Taipei City Dept. of Urban Development**

Summer Intern, Housing Development Section

- Rental-subsidy eligibility checks: cross-matched property, income and insurance records from the tax bureau, land registry and NHI against legal thresholds; handled frontline public inquiries and disputes.

2021 – 2024 — **Tunghai University Student Housing / GEPT**

Resident Assistant (2 yrs) / Exam Staff (part-time)

- Oversaw safety and order in a 100-resident dormitory, mediating disputes and tracking facility safety issues with the dorm supervisor; as part-time exam staff, handled test-paper check-in with zero errors.

Research & Industry

Long-Life Li Battery Evaluation for Smart Water Meters

Partner: a smart-water-meter battery maker **Project lead**

Database & SOP
Established the test database and standardized high-throughput screening procedures (SOP).

Training & tech transfer
Trained QA and R&D teams to independently run large-batch screening and data interpretation.

Nature Materials Journal submission — battery thermal stability study Status **Under review**
Co-author Responsible for TGA thermal measurement

Conferences & Training

Presenter **2026 TwIChE Annual Meeting**
2026/12/05–06 • Feng Chia University
Title: *Weakly Solvating Allyl Ether Electrolytes for Subzero Li ‖ SPAN Batteries: Solvation Structure, Transport Properties, and Separator Wettability*

Vendor seminar **TA Instruments ARES-G3 launch & applications forum**
2026/07/17 • National Taiwan University

Vendor lecture **TA Instruments TGA lecture**
2026/06/03 • National Taiwan University

EHS Licenses

- Dedicated Wastewater and Sewage Treatment Specialist (Class B)
  2026 / 06 · certified · No. GB040231
- Indoor Air Quality Maintenance Management Specialist
  2026 / 08 · certified · No. IA020236
- Waste Disposal Technician (Class A)
  2026 / 10 · certified · No. HA160629
- Occupational Safety and Health Management Personnel (Level B)
  2026 / 11 · mid-Nov exam scheduled
- Occupational Safety Management Specialist (Level A)
  2027 / 03 · exam scheduled

Education

2025 – 2027/08 (expected)
National Taiwan University of Science and Technology
M.S. in Chemical Engineering

Focus on low-temperature electrolytes and battery aging. Coursework: Nanomaterials and Applications (lithography, nanoimprint & self-assembly), Process Safety & Radiation Protection.

2020 – 2024
Tunghai University
B.S. in Chemical & Materials Engineering

Focus on unit operations & process control (Aspen Plus V14 / Aspen Dynamics simulation).

Skills & Instruments

EHS fundamentals

Chemical labelling rules (GHS) · Explosive limits (LEL / UEL) · Occupational exposure limits · Ventilation

Process & operations

Lithography (nanoimprint / self-assembly) · Equipment procurement & maintenance

Software

Aspen Plus V14 · Aspen Dynamics · Microsoft Office · OriginPro · TA Universal Analysis · TA TRIOS

Instruments

Maintenance
TGA Q500 · DSC 2500

Operation
Anton Paar DMA 5002/Lovis 2001 · ARES-G3 · Glovebox · Synchrotron (TLS 20A / TPS 27A / TPS 45A)

Procurement
Glovebox O_2 sensor · DSC sample press · Hydraulic electrode punch · Vacuum pump · Eppendorf Pipette · Low-temperature bath